In [65]:
```
import numpy as np
import pandas as pd
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score,classification_report
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
```

In [66]:
```
iris=load_iris()
X=iris.data[:,:2]     #use first two features
y=(iris.target == 0).astype(int)     #setosa=1,others=0
```

In [67]:
```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

In [68]:
```
model=SVC(kernel='linear',C=1.0)
model.fit(X_train,y_train)
y_pred=model.predict(X_test)
```

In [69]:
```
accuracy=accuracy_score(y_test,y_pred)
print("Accuracy= ",accuracy)
print("\n---Classification Report----")
print(classification_report(y_test,y_pred,target_names=["Setosa","Non-setosa"]))
```

```
Accuracy=  1.0

---Classification Report----
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        20
  Non-setosa       1.00      1.00      1.00        10

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```

In [70]:
```
w=model.coef_[0]
b=model.intercept_[0]
print("Weight vector: ",w)
print("Bias: ",b)
```

```
Weight vector:  [-2.44999622  2.45000378]
Bias:  5.409971273612131
```

In [71]:
```
margin=2/np.linalg.norm(w)
print("Margin: ",margin)
```

```
Margin:  0.577230025457719
```

In [72]:
```
#plot decision boundary and margins

x_min,x_max=X[:,0].min()-1,X[:,0].max()+1
```

```
y_min,y_max=X[:,1].min()-1,X[:,1].max()+1

xx,yy=np.meshgrid(np.linspace(x_min,x_max,200),np.linspace(y_min,y_max,200))
```

In [73]:

```
#plt.contourf(xx,yy,alpha=0.2,cmap='coolwarm')

#plot data points
plt.scatter(X[:,0],X[:,1],c=y,cmap='coolwarm',edgecolors='black')

w=model.coef_[0]
b=model.intercept_[0]

x_line=np.linspace(x_min,x_max,200)
y_line=-(w[0]*x_line+b)/w[1]

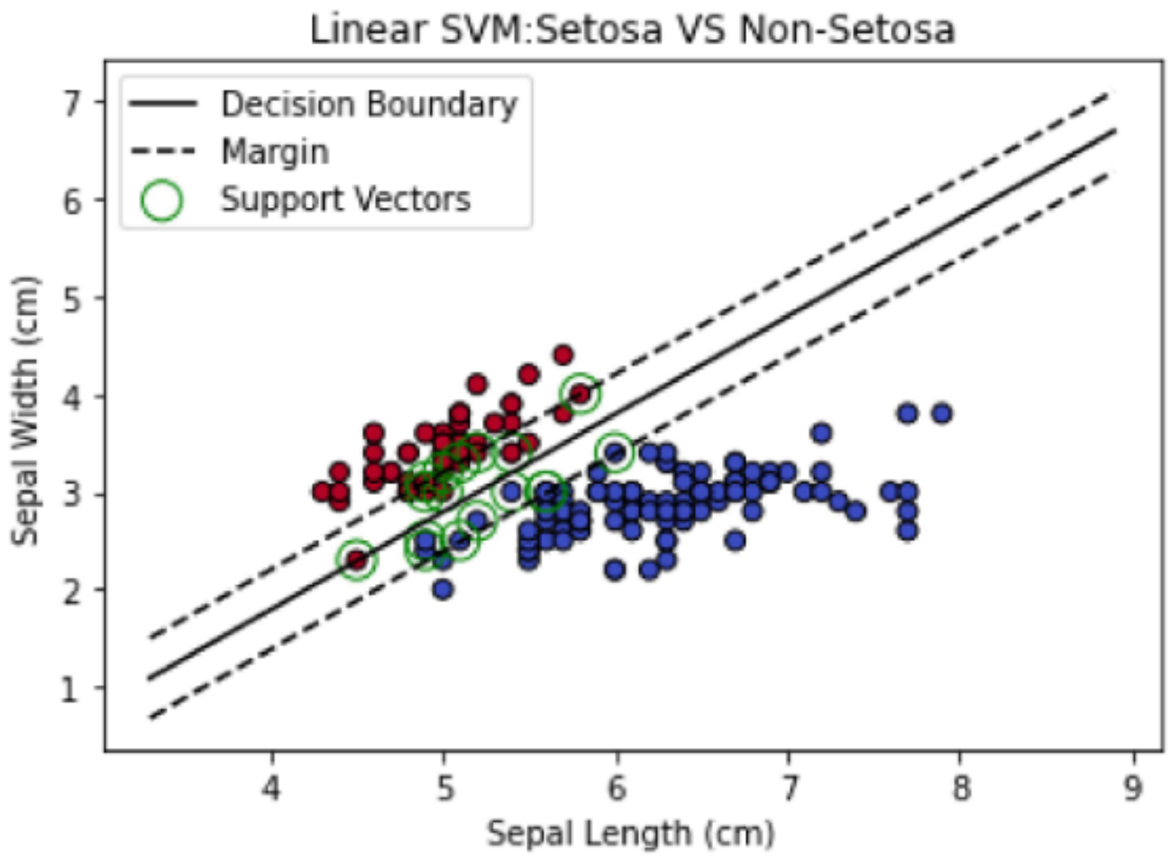
plt.plot(x_line,y_line,'k-',label='Decision Boundary')

plt.plot(x_line,y_line+1/w[1],'k--',label='Margin')
plt.plot(x_line,y_line-1/w[1],'k--')

#highlight support vectors

plt.scatter(
    model.support_vectors_[:,0],
    model.support_vectors_[:,1],
    s=150,facecolors='none',edgecolors='green',
    label='Support Vectors')

plt.xlabel('Sepal Length (cm)')
plt.ylabel('Sepal Width (cm)')
plt.title('Linear SVM:Setosa VS Non-Setosa')
plt.legend()
plt.show()
```



In [ ]: